

South African Inflation Forecasting Project – Full Explanation

1. Project Goal

This project builds a simple econometric model to forecast South African inflation using historical macroeconomic data. The model uses three main economic indicators: inflation (from the CPI index), the interest rate, and the USD/ZAR exchange rate. The objective is to show how past economic data can help explain and forecast future inflation.

2. Tools Used

Python was used for the entire project. The main libraries were: pandas (data manipulation), numpy (numerical calculations), matplotlib (visualization), statsmodels (econometric regression), and sklearn (model evaluation metrics). Jupyter Notebook was used to run the analysis step by step.

3. Data Sources

Economic data was collected from the Federal Reserve Economic Data (FRED) database. The series used were the South African CPI index, the short-term interest rate, and the USD/ZAR exchange rate. These series were downloaded as CSV files and loaded into Python using pandas.

4. Loading Data

```
def load_fred_series(series_id, value_name):
    url = f"https://fred.stlouisfed.org/graph/fredgraph.csv?id={series_id}"
    df = pd.read_csv(url)
    df.columns = ["date", value_name]
    df["date"] = pd.to_datetime(df["date"])
    df[value_name] = pd.to_numeric(df[value_name], errors="coerce")
    return df
```

This function downloads economic data directly from FRED and converts it into a clean pandas dataframe.

5. Merging the Data

```
df = pd.merge(inflation, interest, on="date", how="inner")
df = pd.merge(df, exchange, on="date", how="inner")
```

The three datasets were merged into one table using the date column so all economic indicators align by time.

6. Creating Inflation Rate

```
df["inflation_rate"] = df["inflation_index"].pct_change() * 100
```

The CPI index measures price levels, not inflation itself. Inflation is calculated as the percentage change in the CPI index from one period to the next.

7. Creating Lag Variables

```
df["inflation_lag1"] = df["inflation_rate"].shift(1)
df["inflation_lag2"] = df["inflation_rate"].shift(2)
df["interest_lag1"] = df["interest_rate"].shift(1)
df["usd_zar_lag1"] = df["usd_zar"].shift(1)
```

Lag variables represent past values of economic variables. Economists often use lagged variables because current inflation is often influenced by past inflation and past economic conditions.

8. Running the Regression Model

```
X = df[["inflation_lag1", "inflation_lag2", "interest_lag1", "usd_zar_lag1"]]
y = df["inflation_rate"]

X = sm.add_constant(X)
model = sm.OLS(y, X).fit()
```

This regression estimates how lagged inflation, interest rates, and exchange rates influence current inflation.

9. Predicting Inflation

```
df["predicted_inflation"] = model.predict(X)
```

The regression model produces predicted values for inflation based on the estimated relationships.

10. Model Evaluation

```
mae = mean_absolute_error(df["inflation_rate"], df["predicted_inflation"])
rmse = mean_squared_error(df["inflation_rate"], df["predicted_inflation"]) ** 0.5
```

Two metrics were used to evaluate the model: Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). These measure the average forecasting error.

11. Visualization

```
plt.plot(df["date"], df["inflation_rate"], label="Actual Inflation")
plt.plot(df["date"], df["predicted_inflation"], label="Predicted Inflation")
plt.legend()
```

The final graph compares real inflation with model predictions. The model captures the general trend but smooths out short-term spikes.

12. Conclusion

This project demonstrates how economic data can be used to build simple forecasting models. While the model captures general inflation trends, it cannot fully explain sudden economic shocks. More advanced models or additional variables could improve forecasting accuracy.